

An OpenUSD Production Pipeline with Very Little Coding

Empowering 3D artists with a parallel workflow using off-the-shelf software.

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ABSTRACT

We present a streamlined animation production pipeline leveraging OpenUSD and Houdini’s procedural strengths to enable collaborative, parallel workflows. Our approach minimizes coding requirements, empowering artists to iterate simultaneously on shared assets and shots for real-time project visualization. Developed and proven in an academic setting, this pipeline demonstrates adaptability and scalability for small studio environments, successfully fostering iterative workflows among users.

CCS CONCEPTS

• **Computing methodologies** → Computer graphics; Graphics systems and interfaces.

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1 INTRODUCTION

Can a modern, OpenUSD-driven 3D animation pipeline be created without extensive coding resources?

Our undergraduate animation program faced this challenge as student teams scaled to 20+ members on their annual short film project. To facilitate collaboration and iteration, we experimented with building an OpenUSD-based solution. The school lacked the resources for heavy custom coding.

Our goal was to automate tedious tasks: generating turntables for review, assembling asset variants into USD files, and creating layer stacks with the latest asset versions. While Python coding with the OpenUSD library was possible, it was overly demanding. Instead, we leveraged the procedural power of Houdini’s stage context. Template scenes were designed for each task, requiring minimal Python code to launch Houdini, execute procedures, and integrate results.

2 MODERN WORKFLOW

Traditional VFX and animation pipelines followed a linear ‘assembly line’ model, hindering collaboration. Modern workflows embrace

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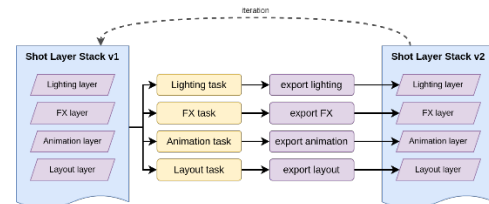


Figure 1: Diagram of the parallel and iterative nature of our pipeline, showing only some of the shot departments. The scene file is assembled by providing the OpenUSD engine with layers. This file is being loaded by several departments at the same time to produce a new version of each department’s layer. Then the layers can be assembled again to repeat the process.

a parallel approach, where artists across departments contribute simultaneously to assets and shots. The key is for artists to import their colleagues’ work, add their own contributions, and then export only those updates.

Legacy software like Maya and 3ds Max weren’t designed for this. Studios are often required to code custom scene builders and file formats in order to assemble the result of upstream departments in the pipeline. OpenUSD simplifies parallel workflows with its sparse data structure and fast composition. However, extensive coding can still be necessary for standardization of the assets and shots structures. Our goal was to investigate how far we could push a parallel workflow with minimal coding, while still retaining uniformity in file structures.

3 OVERVIEW

3.1 Purposes and variants

To ensure flexibility, we designed a software-agnostic method for packaging mesh resolutions and variants into single USD files with metadata (purpose, kind, composition). Artists export work as usd format to designated folders. File names detail asset name, work type (geometry, texture, etc.), variant, and resolution. This unified process works across modeling, animation, and FX. Right-clicking the folder triggers a script. It launches Houdini in batch mode with the correct template scene. Houdini builds a hierarchy, adds metadata, and creates a department-based variant set. The result is one .usda file per department for each asset/instance. This powerful solution requires only 600 lines of code, thanks to Houdini’s capabilities.

3.2 The shot/asset layer stack

At any time, users can easily assemble an asset or shot to view the latest work across departments by triggering a script. It analyzes the existing layout layer and references the most recently published files from each department to the right spot in the hierarchy. Department layers are then combined into a single layer stack file using a predefined order. This script is remarkably compact (only a few hundred lines of code) and can be repeatedly run to generate updated views of ongoing work.

3.3 The turn-table

We automated the tedious task of turntable creation. Artists trigger a script from an asset folder, and Houdini generates a standardized scene with calibrated lighting and a rotating animation. This system

adapts to assets of any size and allows for department-specific materials.

4 RESULTS

This pipeline was successfully deployed in the production of two animated films, created by teams of 22 and 30 students. The pipeline empowered students to iterate rapidly across all production aspects, facilitating frequent full-film reviews incorporating the combined work of all departments. We strongly believe this pipeline enhanced the visual quality of the final films while simultaneously preparing students for industry realities, where collaboration with hundreds of individuals is the norm. We also believe that a similar pipeline can be built in a small animation studio with minimal effort and that it would provide the same benefits.